

DSA0601 • DATA HANDLING AND VISUALIZATION LAB EXPERIMENT RECORD

Set 15: Product Sales Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyze monthly product sales data using a grouped bar chart, a sales trend chart and a tabular representation, and to develop a Tableau dashboard combining the visualizations for interactive exploration of sales data.

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

PROGRAM 1: Grouped Bar Chart of Product Sales

Create the monthly sales data and visualize the sales of each product for the first quarter. Label the chart elements.

Program 1: Grouped Bar Chart for Product Sales

```
product <- c("Product A", "Product B", "Product C")
```

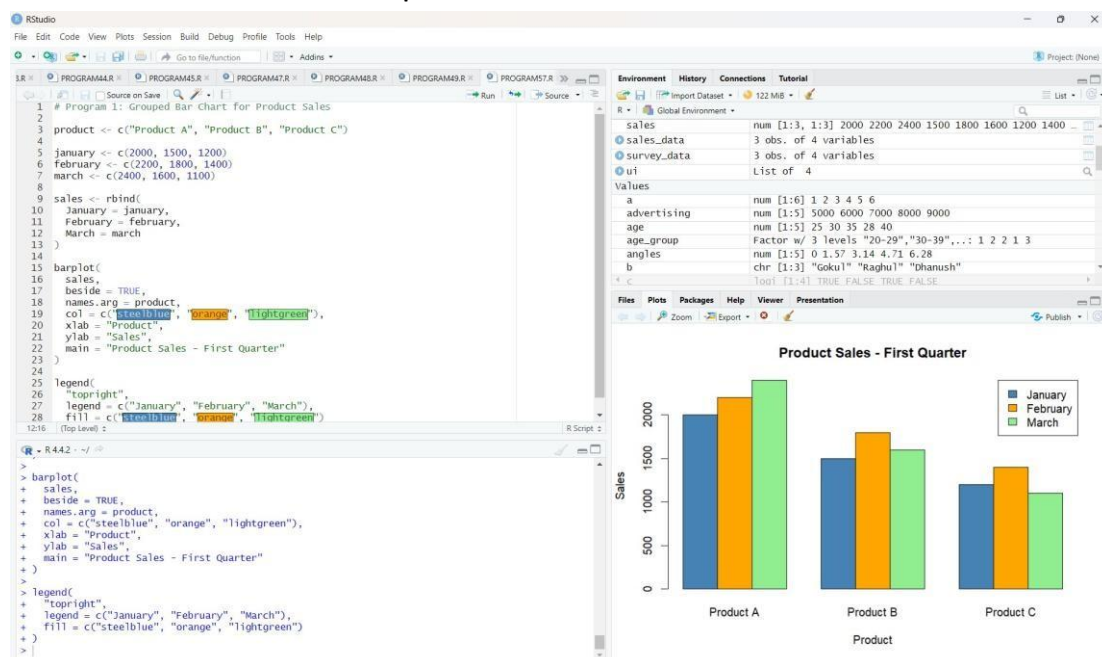
```
january <- c(2000, 1500, 1200)
february <- c(2200, 1800, 1400)
march <- c(2400, 1600, 1100)
```

```
sales <- rbind(
  January = january,
  February = february,
  March = march
)
```

```
barplot(sales,
  beside = TRUE,
  names.arg = product,
  col = c("steelblue", "orange", "lightgreen"),
  xlab = "Product", ylab = "Sales",
  main = "Product Sales - First Quarter"
)
```

```
legend("topright", legend = c("January",
  "February", "March"), fill = c("steelblue",
  "orange", "lightgreen")
)
```

Output 1: Product Sales - First Quarter



Observation: Product A has the highest sales in all three months, rising from 2000 to 2400, while Product C has the lowest sales in every month.

PROGRAM 2: Monthly Product Sales Trend

Generate a chart to represent the sales trend of all products over the three months. Label the chart elements.

```
# Program 2: Monthly Product Sales Trend month
<- c("January", "February", "March")

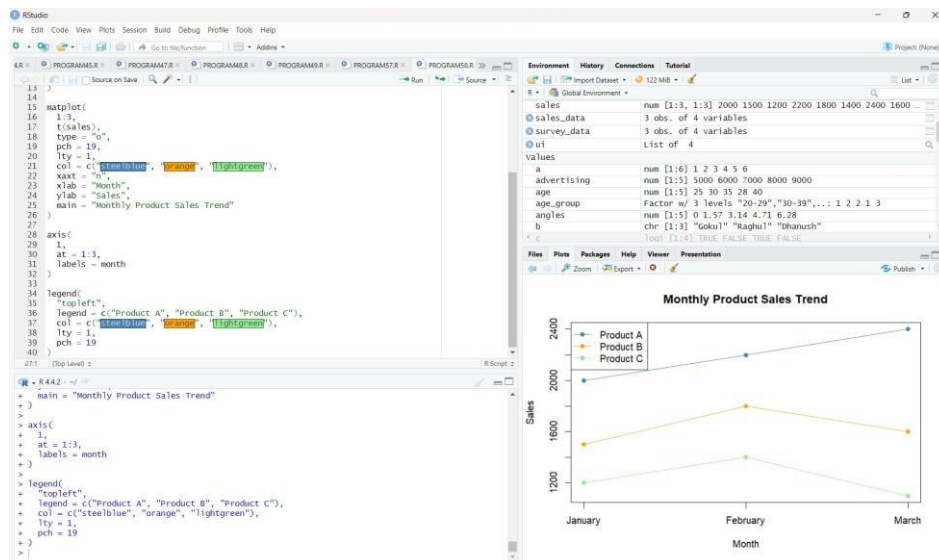
sales <- cbind(
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)

matplot(
  1:3,
  t(sales),
  type = "o",
  pch = 19,
  lty = 1,
  col = c("steelblue", "orange", "lightgreen"),
  xaxt = "n", xlab = "Month", ylab = "Sales",
  main = "Monthly Product Sales Trend"
)

axis( 1,
  at = 1:3,
  labels = month
)

legend( "topleft", legend = c("Product A",
"Product B", "Product C"), col = c("steelblue",
"orange", "lightgreen"), lty = 1, pch = 19
)
```

Output 2: Monthly Product Sales Trend



Observation: Product A shows steady growth, while Products B and C rise in February and then fall in March. The overall trend is shown as a stacked area chart in the Tableau dashboard.

PROGRAM 3: Product Sales Table

Build a table to show the monthly sales data for each product. Label the table elements.

Program 3: Product Sales Table

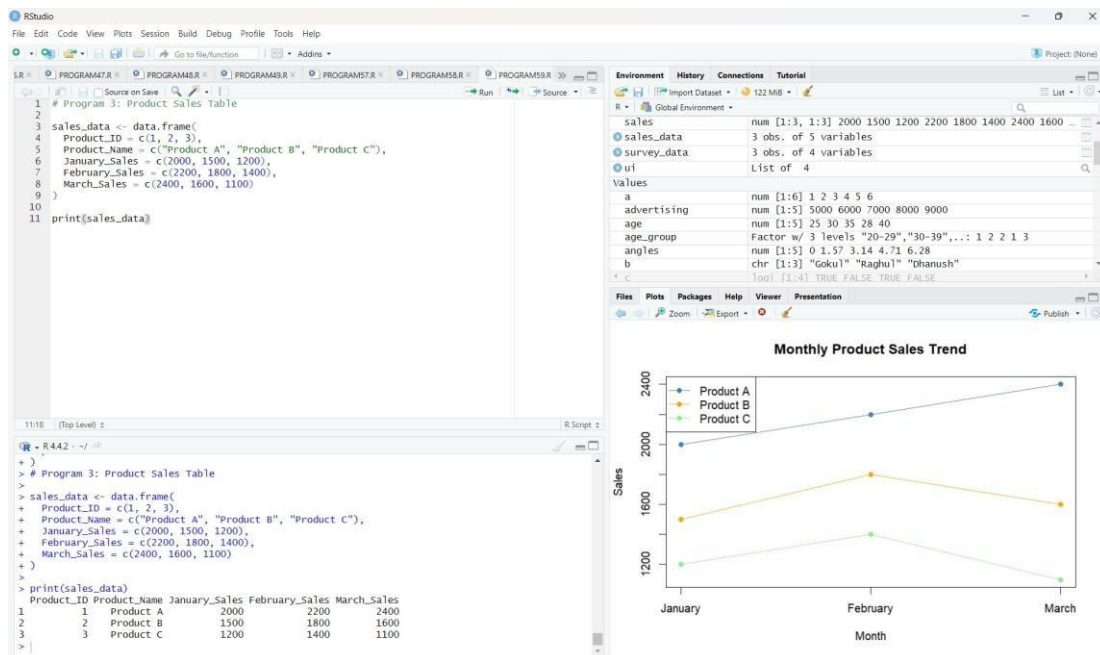
```

sales_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)

print(sales_data)

```

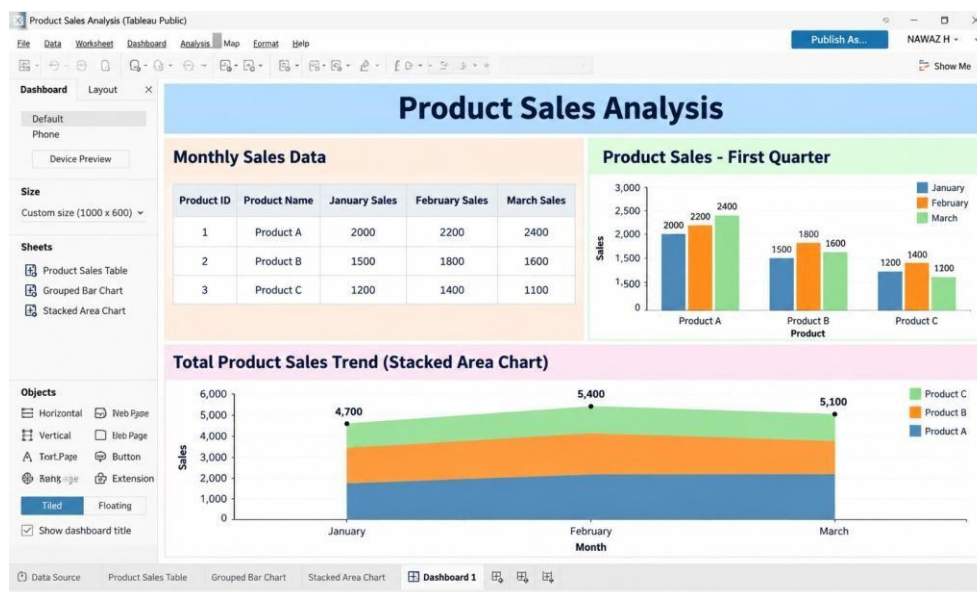
Output 3: Product Sales Table



Observation: The table displays the product ID, product name, and the January, February, and March sales of the three products.

PROGRAM 4: Tableau Dashboard

Develop a Tableau dashboard combining the product sales table, grouped bar chart, and stacked area chart for interactive exploration of sales data.



Output 4: Tableau Product Sales Analysis Dashboard

Observation: The Tableau dashboard combines the monthly sales table, the grouped bar chart, and the stacked area chart. The total sales are 4,700 in January, 5,400 in February, and 5,100 in March.

Procedure

1. Create the product and monthly sales vectors in R.
2. Combine the monthly sales into a matrix using `rbind()` and `cbind()`.
3. Use `barplot()` with `beside = TRUE` to generate the grouped bar chart.
4. Use `matplot()`, `axis()`, and `legend()` to plot and label the monthly sales trend.
5. Create and print the product sales table using `data.frame()`.
6. Enter the sales data into Tableau and create the Product Sales Table, Grouped Bar Chart, and Stacked Area Chart worksheets.
7. Combine the table, grouped bar chart, and stacked area chart into a dashboard titled Product Sales Analysis.

Result

The monthly product sales data was successfully visualized using a grouped bar chart, a sales trend chart, and a table in R. A Tableau dashboard was also created by combining the sales table, grouped bar chart, and stacked area chart for interactive exploration.

Key R Functions Used

Function	Purpose
<code>c()</code>	Creates the product and monthly sales vectors.
<code>rbind()</code> / <code>cbind()</code>	Combines the monthly sales into a matrix.
<code>barplot()</code>	Creates the grouped bar chart.
<code>matplot()</code>	Plots the monthly sales trend for all products.
<code>axis()</code>	Labels the months on the x-axis.
<code>legend()</code>	Labels the products or months in the chart.
<code>data.frame()</code>	Creates the product sales table.